

BUSINESS INSIGHTS REPORT

Bank Customer Churn Analysis

A comprehensive analysis of churn drivers, customer segments, predictive models, and retention recommendations

Dataset: 10,000 customers | Analysis Date: 2026

01 Executive Summary

This report presents findings from a full-cycle data analysis of bank customer churn, covering exploratory analysis, machine learning prediction, customer segmentation, and interactive visualisation. The objective is to equip business leaders and relationship managers with clear, actionable intelligence to reduce churn and improve customer lifetime value.

10,000 Customers Analysed	20.4% Overall Churn Rate	2,037 Customers Lost	7,963 Customers Retained
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One in five customers has left the bank. At an industry average customer acquisition cost of \$200–\$400, this represents a potential revenue leakage of \$400K–\$800K on replacement costs alone — before accounting for lost lifetime value. The analysis identifies three primary churn drivers: geography (Germany), age (51–60 segment), and product over-concentration (3–4 products). With targeted intervention, an estimated 40–60% of at-risk churners are recoverable.

At-a-Glance Findings

Dimension	Highest Risk	Churn Rate	Lowest Risk	Churn Rate
Geography	Germany	32.4%	France	16.2%
Age Group	51–60	56.2%	18–30	7.5%
No. of Products	3–4 Products	83–100%	2 Products	7.6%
Gender	Female	25.1%	Male	16.5%
Membership	Inactive	26.9%	Active	14.3%
Segment	High-Risk	33.3%	Loyal	9.3%

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Data Overview & Methodology

The analysis uses a dataset of 10,000 bank customers with 14 variables covering demographics, product usage, financial behaviour, and churn outcome. Four analytical modules were executed in sequence.

Module	Technique	Primary Output
Module 1 — EDA	Correlation, distribution analysis	Churn driver identification
Module 2 — Prediction	Logistic Regression, Random Forest, XGBoost	Model accuracy & AUC-ROC scores
Module 3 — Segmentation	K-Means clustering (k=4), PCA	4 customer segments with profiles
Module 4 — Visualisation	Matplotlib, Seaborn, Plotly	Static + interactive dashboards

Dataset Variables

Variable	Type	Description
CreditScore	Numeric	Customer credit score (350–850)
Geography	Categorical	Country: France, Germany, Spain
Gender	Categorical	Male / Female
Age	Numeric	Customer age in years
Tenure	Numeric	Years as a bank customer
Balance	Numeric	Current account balance (\$)
NumOfProducts	Numeric	Number of bank products held (1–4)
IsActiveMember	Binary	Actively uses banking services (1=Yes)
EstimatedSalary	Numeric	Estimated annual salary (\$)
Exited	Binary (Target)	Churned = 1 Retained = 0

03 Exploratory Data Analysis — Key Findings

3.1 Geographic Breakdown

Geography is the strongest single demographic predictor of churn. German customers churn at exactly twice the rate of French and Spanish customers, despite representing only 25% of the customer base.

Country	Total Customers	Churned	Churn Rate	Share of Base
France	5,014	810	16.2%	50.1%
Germany	2,509	814	32.4%	25.1%
Spain	2,477	413	16.7%	24.8%

- Germany produces 40% of all churned customers while being 25% of the base — a 1.6× over-representation.
- France and Spain are statistically equivalent in churn behaviour (16.2% vs 16.7%).
- German operations warrant immediate investigation: pricing competitiveness, product-market fit, and service quality metrics should be reviewed.

3.2 Age Group Analysis

Age is the most powerful predictor of churn after product count. The 51–60 cohort has a churn rate of 56.2% — more than half of all customers in this bracket leave the bank. This suggests a life-stage misalignment where the bank's product offering fails to meet evolving financial needs.

Age Group	Total Customers	Churned	Churn Rate	Risk Level
18–30	1,946	146	7.5%	Low
31–40	4,451	538	12.1%	Low–Medium
41–50	2,320	788	34.0%	High
51–60	797	448	56.2%	Critical
60+	464	115	24.8%	Medium

- 51–60 is the single highest-risk cohort: 56.2% churn, affecting nearly 800 customers.
- The 41–50 bracket, while lower at 34%, represents the largest volume of churners (788) due to its bigger population.
- Young customers (18–30) are highly loyal at 7.5% churn — the bank's strongest relationship-building window.

3.3 Product Count Effect

The relationship between product count and churn is non-linear and counter-intuitive. While 2 products yields the lowest churn (7.6%), adding a third product dramatically increases churn risk to 82.7%, and customers with 4 products have a 100% churn rate.

Products Held	Total Customers	Churned	Churn Rate	Insight
1 Product	5,084	1,409	27.7%	High single-product dependency
2 Products	4,590	348	7.6%	Optimal sweet spot
3 Products	266	220	82.7%	Critical over-concentration
4 Products	60	60	100.0%	Complete churn — all left

- The 2-product zone is where the bank retains customers best. This is the retention target for all segments.
- 3–4 product customers likely feel over-sold or financially overextended. Product simplification is urgent.
- All 60 customers with 4 products have already churned — this cohort requires retrospective analysis.

3.4 Gender & Membership Activity

Female customers churn at a 52% higher rate than male customers (25.1% vs 16.5%). Inactive members are nearly twice as likely to churn as active ones. Neither credit score, salary, nor tenure show meaningful differences between churned and retained groups.

Factor	Group	Churn Rate	Implication
Gender	Female	25.1%	Targeted retention products for female customers
Gender	Male	16.5%	Lower risk — maintain current engagement
Activity	Inactive	26.9%	Re-engagement campaigns are high ROI
Activity	Active	14.3%	Activity = loyalty; reward active behaviour
Credit Score	Churned vs Retained	645 vs 652	Negligible difference — not a useful predictor alone
Balance	Churned vs Retained	\$91K vs \$73K	Churned customers have more wealth elsewhere

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Machine Learning — Churn Prediction

Three classification models were trained on 80% of the dataset (8,000 customers) and evaluated on the remaining 20% (2,000 customers). Model performance was measured using Accuracy and AUC-ROC, with AUC-ROC being the primary metric as it accounts for class imbalance.

4.1 Model Performance Summary

Model	Accuracy	AUC-ROC	Best Use Case	Recommended?
Logistic Regression	~81%	~0.77	Baseline / explainability	Baseline only
Random Forest	~86%	~0.86	Balanced accuracy & speed	Yes
XGBoost	~87%	~0.88	Highest AUC — production use	Yes — Primary

- XGBoost is the recommended production model with the highest AUC-ROC (~0.88), meaning it correctly ranks 88% of churner-vs-retainer pairs.
- Random Forest is a strong secondary model, more interpretable and nearly as accurate.
- Logistic Regression serves as a performance baseline — any model not exceeding its AUC would be discarded.

4.2 Top Predictive Features

Both tree-based models (Random Forest and XGBoost) consistently rank the same variables as the most important predictors of churn, confirming the EDA findings.

Rank	Feature	Importance	Business Interpretation
1	Age	Highest	Older customers at significantly higher risk
2	NumOfProducts	Very High	3–4 products = near-certain churn trigger
3	IsActiveMember	High	Inactivity is a leading indicator of departure
4	Balance	Medium	Higher-balance customers more likely to leave
5	Geography	Medium	German customers systematically more at-risk
6	Gender	Lower	Female customers slightly more likely to churn
7	CreditScore	Low	Minimal predictive power in isolation
8	Tenure	Low	Length of relationship not protective enough

The model can be operationalised as a real-time churn propensity score, flagging at-risk customers before they leave. Customers scoring above a 0.6 probability threshold should trigger an automated retention workflow.

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Customer Segmentation

K-Means clustering (k=4, validated by elbow method) combined with business rules identified four distinct customer segments. Each segment has a unique risk profile, financial footprint, and recommended retention strategy.

Segment	Customers	% of Base	Churn Rate	Avg Balance	Active Rate
Loyal	557	5.6%	9.3%	~\$105K	100%
Dormant	3,617	36.2%	13.8%	\$0	~45%
Moderate-Risk	2,635	26.4%	16.1%	~\$119K	~55%
High-Risk	3,191	31.9%	33.3%	~\$120K	0%

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Retention Strategy Recommendations

The following recommendations are derived from combining EDA insights, model feature importance, and segment profiles. Actions are prioritised by churn risk and estimated impact.

HIGH-RISK SEGMENT

 | Churn Rate: 33.3% churn

3,191 inactive customers or those holding 3+ products. Highest churn rate of all segments. Average balance ~\$120K — high-value customers actively considering departure.

- Immediate outreach: personal banker call within 48 hours for all inactive accounts
- Re-engagement offer: fee waiver or bonus interest rate for 3 months
- Product review: simplify portfolios — guide 3–4 product holders to 2 optimal products
- Loyalty incentives: cashback or points programme to rebuild engagement
- Predictive trigger: automate a retention workflow when IsActiveMember flips to 0

DORMANT SEGMENT

 | Churn Rate: 13.8% churn

3,617 customers with zero balance — the largest single segment at 36% of the base. They have salary (~\$99K avg) but are not depositing with this bank. Silent attrition risk is high.

- Deposit incentive: promotional interest rate (e.g. 5% for 6 months) to fund accounts
- Salary crediting campaign: payroll deposit bonus to create recurring engagement
- Mobile app push: personalised nudges showing savings goals and milestones
- Cross-sell: entry-level investment product aligned to their salary bracket
- Win-back email series: reinforce account value and dormancy benefits

 MODERATE-RISK SEGMENT | Churn Rate: 16.1% churn

2,635 engaged customers with 1 product but not maximising their banking relationship. Average balance ~\$119K. Churn is preventable with targeted deepening of the relationship.

- Product upsell: introduce credit card or savings account to reach the 2-product sweet spot
- Financial health check: free annual review to build trust and stickiness
- Milestone rewards: tenure-based perks at 3, 5, and 10-year relationship anniversaries
- Digital engagement: enrol in mobile banking features — budgeting tools, analytics
- Referral programme: reward existing customers for bringing in new accounts

 LOYAL SEGMENT | Churn Rate: 9.3% churn

557 premium customers — active members with 2 products and credit score above 650. The bank's most stable cohort. Priority is to protect, reward, and leverage as brand ambassadors.

- VIP programme: dedicated relationship manager and priority service channel
- Exclusive rates: preferential mortgage and loan rates to cement long-term retention
- Advocacy programme: referral bonuses — loyal customers are the best brand ambassadors
- Premium upgrade path: offer premium or black card tier with additional benefits
- Early access: beta product features, exclusive events, and new product launches

07 **Priority Action Matrix**

The following matrix maps the highest-impact actions by effort and expected reduction in churn rate. Items in the top-left quadrant (high impact, low effort) should be executed first.

Priority	Action	Target Segment	Expected Impact	Effort
1 — Immediate	Automate churn propensity scoring	All	Flag 80%+ of churners before departure	Low
2 — Immediate	Personal outreach to inactive accounts	High-Risk	15–25% re-engagement rate	Medium
3 — 30 days	Reduce 3–4 product customers to 2	High-Risk	Churn rate down from 83% toward 8%	Medium
4 — 30 days	Germany-specific retention campaign	All German	Close 16-point gap vs France	Medium
5 — 60 days	Age 51–60 financial planning programme	Moderate-Risk	Reduce 56% churn rate by est. 20pts	High
6 — 60 days	2-product cross-sell for single-product	Moderate-Risk	Reduce churn from 27.7% toward 7.6%	Low
7 — 90 days	Deposit activation campaign for Dormant	Dormant	Convert 10–15% of zero-balance accounts	Medium
8 — Ongoing	VIP programme for Loyal segment	Loyal	Protect \$55M+ in relationship value	Low

08 Conclusion

This analysis establishes a clear, evidence-based picture of churn at this bank. The 20.4% churn rate is not uniformly distributed — it is concentrated in identifiable, addressable pockets: German customers, customers aged 51–60, those with 3 or more products, and inactive members.

The machine learning models achieve an AUC-ROC of up to 0.88, enabling predictive identification of at-risk customers before they leave. Combined with the four-segment framework — High-Risk, Dormant, Moderate-Risk, and Loyal — the bank has the tools to move from reactive to proactive retention.

The single most impactful intervention is guiding customers to the 2-product relationship, where churn falls to 7.6%. If the bank could move just 20% of its single-product base to 2 products, an estimated 300–400 additional customers would be retained annually.

Key Takeaway for Leadership

Churn is predictable, segmentable, and reducible. The data shows exactly where the bank is losing customers and why. The next step is translating these insights into operational workflows — a churn propensity score embedded in CRM, segment-based outreach campaigns, and a product simplification initiative for over-concentrated customers. A 5-point reduction in overall churn rate (from 20.4% to 15.4%) would retain approximately 500 additional customers per year.

A Appendix — Output Files

File	Module	Description
churn_bank_customers.xlsx	Source	Raw dataset — 10,000 customers, 14 variables
exploratory_data_analysis.py	Module 1	EDA: distributions, correlations, churn breakdowns
churn_prediction.py	Module 2	ML models: Logistic Regression, Random Forest, XGBoost
customer_segmentation.py	Module 3	K-Means clustering, PCA, segment profiling
visualisation.py	Module 4	Static dashboard (Matplotlib) + interactive (Plotly)
confusion_matrices.png	Module 2	Confusion matrix comparison — all 3 models
roc_curves.png	Module 2	ROC curve comparison with AUC scores
model_comparison.png	Module 2	Accuracy and AUC bar chart
feature_importance.png	Module 2	Random Forest and XGBoost feature rankings
elbow_plot.png	Module 3	K-Means elbow method — optimal k selection
segmentation_dashboard.png	Module 3	6-panel segmentation visualisation
customers_segmented.xlsx	Module 3	Full dataset with Cluster and Segment labels
churn_dashboard_static.png	Module 4	5-chart static churn dashboard
churn_dashboard_interactive.html	Module 4	Interactive Plotly dashboard — open in browser